

Course code: CSE 422
Course name: Coding Practice
Type of Exam: Opened book / Practice on Computer

Exam code: 01
Duration: 120'
Maximum mark: 100

Instructions:

- *Non-programming calculator: Yes ☒ | No ☐*
- *This paper consists of 5 questions. This paper carries 100 marks:*
Question 1 (20 marks).
Question 2 (40 marks).
Question 3 (40 marks).

Students must create the project using the naming convention:

Submission: Students zip the source code, report file in one compressed file for submission.

Format: DateNow(DD-MM-YYYY)_CP_YourName

Example: 03092025_CP_NguyenMinhThien

Students are allowed to use PDF and printed documents. However, the internet, personal computers, and cell phones are not allowed.

Website allow: <https://app.diagrams.net/>

Question 1: Coding Style and Conventions (20 marks)

Task:

You are given the following C# class:

```
class user
{
    string name;
    int age;

    public user(string Name, int Age)
    {
        name = Name;
        age = Age;
    }
}
```

Requirements:

1. Identify and explain the coding style and convention issues in the above code.
2. Rewrite the class following C# best practices.
3. Save your rewritten code and your explanation in a file named **Question1.cs** inside your folder.

Question 2: Object-Oriented Programming and SOLID Principles (40 marks)

Task:

You are developing a **Payment Processing System** that supports multiple payment methods. The system should:

- Follow the **Open/Closed Principle (OCP)**, allowing new payment methods to be added without modifying existing code.
- Use **Dependency Injection (DI)** to decouple the payment processing logic.

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- Support at least **three payment methods: Credit Card, PayPal, and Bank Transfer.**

Requirements:

1. Create a **C# console project** to implement the system.
2. Design an interface **IPaymentMethod** to define a **ProcessPayment(decimal amount)** method.
3. Implement at least **three payment classes** that adhere to **IPaymentMethod**.
4. Develop a **PaymentProcessor** class that accepts any **IPaymentMethod** instance and processes payments.
5. Demonstrate usage in a **Main** method, ensuring flexibility for future expansions.

Question 3: Microservice Architecture Design (40 marks)

Apply the **Microservice architecture model** to design a system and create a detailed system diagram based on the following description.

System Components:

1. **Frontend:**
 - **Web App:** For library staff and administrators.
 - **Mobile App:** For library users (book reservations/borrowing, checking history).
2. **API Gateway:**
 - Orchestrates requests from clients to microservices.
3. **Backend Services (Microservices):**
 - **Authentication & Authorization Service:** Manages users, login, and role-based access control.
 - **Book Management Service:** Manages books (add, edit, delete, search).
 - **Borrow & Return Service:** Manages book borrowing and returns.
 - **User Management Service:** Manages user information.
 - **Notification Service:** Sends notifications when books are due.
4. **Load Balancer:**
 - Ensures load balancing.
5. **Firewall (DMZ):**
 - Protects the system from external threats.
6. **Database:**
 - Each service has its own database.
7. **Monitoring Server:**
 - Monitors system performance.

Requirements:

1. Draw a system architecture diagram illustrating how the components interact.
2. Explain the role of each component in the system.
3. Propose technologies to use for each component in the model

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Lecturer's signature


Nguyen Minh Thien